(Please write your Enrollment Number) Enrollment No. 01601172021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech. CSE-AI, Semester: Fourth
(April -May, 2023)

Subject Code: BAI - 202	Subject: Computer Networks

Time :3 Hours Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5*8=20)

(a) List the characteristics of broadband coaxial cable.
(b) What limitations of a bridge are overcome by a router?
(c) State the purpose of layering in Networks.
(d) List two advantages and two disadvantages of Flooding.
(e) For n devices in a network, what is the number of cable links required for a mesh, ring and star topology, respectively?

(f) What are the network address, broadcast address, and the subnet mask for a host with the Given IP address 183.203.107.171/20

(g) How is IP version 4 address space is divided into different classful addresses. Explain ?
(h) Draw and explain different fields of UDP header.

UNIT-I

Q2 Define fiber optic communication principle and light propagation through fiber optic cable **(10)**

Q3 Compare and contrast ISO-OSI and TCP/IP Model **(10)**

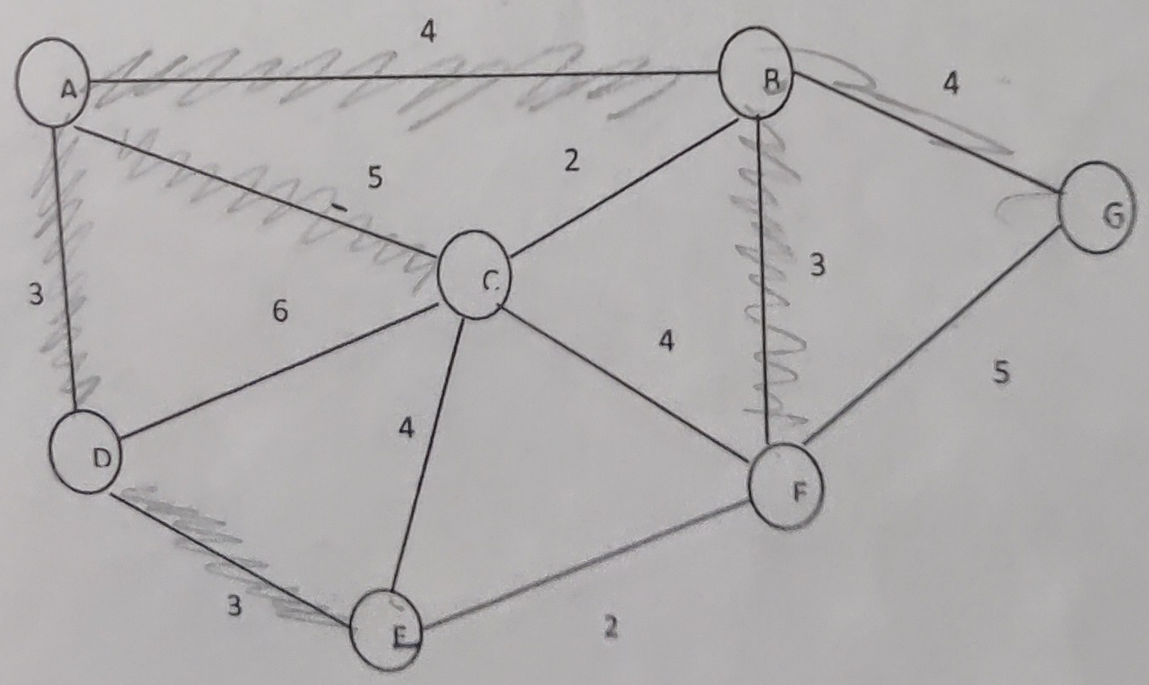
UNIT-II

Q4 A group of terminals are to be connected to a central site using ALOHA schemes over a 9600 bps channel. Frame length is 200 bits. Terminals generate on an average one frame in every two minutes. For maximum throughput condition, estimate the number of terminals that can be supported if the scheme used is (a) Pure ALOHA (b) Slotted ALOHA. **(10)**

Q5 Explain the reason for moving from Stop-and-Wait ARQ Protocol to the Go-Back-N ARQ protocol. **(10)**

UNIT-III

Q6 Using Dijkstra's algorithm to find the shortest path tree for the node A in the figure given below **(10)**



Q7 An organization is granted the block 130.56.0.0/16 The administrator wants to create 1024 nets. **(10)**

(i) Find the subnet mask.
(ii) Find the number of addresses in each subnet.
(iii) Find the first and the last address in the first subnet.
(iv) Find the first and the last address in the last subnet (subnet 1024)

UNIT-IV

Q8 Explain the use of TCP and UDP header Fields ? (10)

Q9 (a) Discuss the features of HTTP and also discuss how HTTP works. (10)
(b) Explain the working of Electronic mail. How SMTP used in Email applications